

Xingran Chen

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Research Interests

My research focuses on the information-theoretic foundations and algorithm design for networked intelligence and security. More recently, I have been developing rigorous theory and algorithms for decentralized learning via random walks, spanning analysis, design, optimization, and real-world applications.

Academic Positions

Assistant Professor	Jul. 2026 - present
Engineering Systems and Design Pillar Singapore University of Technology and Design	
Postdoctoral Researcher	Jan. 2025 - Apr. 2026
Department of Electrical and Computer Engineering Rutgers University Advisor: Prof. Salim El Rouayheb	
Assistant Professor	Sep. 2023 - Apr. 2026
School of Information and Communication Engineering University of Electronic Science and Technology of China Academic leave since Jan. 2025	

Education

Ph.D. in Electrical and Systems Engineering	2018 - 2023
<i>University of Pennsylvania</i> Advisor: Prof. Shirin Saeedi-Bidokhti	
M.A. in Applied Mathematics and Computational Science	2016 - 2018
<i>University of Pennsylvania</i> Thesis Advisor: Prof. Saswati Sarkar	
M.S. in Operations Research	2015 - 2016
<i>Central South University</i> (transfer to Upenn in 2016)	
B.S. in Mathematics and Statistics	2011 - 2015
<i>Central South University</i> (GPA: 4.0/4.0)	

Awards

Annual Outstanding Faculty Award (University of Electronic Science and Technology of China)	2024
IEEE Communications Society & Information Theory Society Joint Paper Award	2023
IEEE INFOCOM Student Conference Award	2021
IEEE ISIT Student Travel Grant	2019
The Dean's Fellowship (University of Pennsylvania)	2018
Ganster Engineering Fellowship (University of Pennsylvania)	2018

Research Grants

Young Scientists Fund of the NSFC	Jan. 2025 - Dec. 2027
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Grant No: 62401111

Title: Freshness-oriented decentralized random access theory and algorithms

Funding Agency: National Natural Science Foundation of China

Role: Solo PI

Fundamental Research Funds for the Central Universities

Sep. 2024 - Aug. 2026

Grant No: ZYGX2024XJ001

Title: Research on Decentralized Random Access Protocols for Information Freshness

Funding Agency: Ministry of Education of China

Role: Solo PI

Publications

Preprints

- [P1] **X. Chen**, R. Bhagat, G. Ayache, R. Bitar, Y. Gong, and S. El Rouayheb, [RW-LoRA: Communication-Efficient Decentralized LoRA Fine-Tuning via Random Walks](#), under review.
 - [P2] Y. Zhuang, K. Yang, **X. Chen***, [Adaptive Requesting in Decentralized Edge Networks via Non-Stationary Bandits](#), under review. (*: corresponding author)
 - [P3] **X. Chen**, P. Parag, R. Bhagat, and S. El Rouayheb, [Self-Creating Random Walks for Decentralized Learning under Pac-Man Attacks](#), under review.
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Journal

- [J1] **X. Chen**, K. Li, and K. Yang, [Timely Requesting for Time-Critical Content Users in Decentralized F-RANs](#), *IEEE Transactions on Networking (ToN)*, 2025.
 - [J2] **X. Chen**, Y. Zhuang, and K. Yang, [Age of Computing: A Metric of Computation Freshness in Communication and Computation Cooperative Networks](#), *IEEE Transactions on Mobile Computing (TMC)*, 2025.
 - [J3] H. Nikpey, J. Kim, **X. Chen**, S. Sarkar, and S. Saeedi Bidokhti, [Group Testing with Correlation under Edge-Faulty Graphs](#), *IEEE Transactions on Information Theory (TIT)*, 2024.
 - [J4] **X. Chen**, H. Nikpey, J. Kim, S. Sarkar, and S. Saeedi-Bidokhti, [Containing A Spread through Sequential Learning: to Exploit or to Explore?](#), *Transactions on Machine Learning Research (TMLR)*, 2023.
 - [J5] J. Kim, **X. Chen**, H. Nikpey, S. Saeedi-Bidokhti and S. Sarkar, [Tracing and Testing Multiple Generations of Contacts to COVID-19 Cases: Cost-Benefit Tradeoffs](#), *Royal Society Open Science*, 2022.
 - [J6] **X. Chen**, K. Gatsis, H. Hassani and S. Saeedi-Bidokhti, [Age of Information in Random Access Channels](#), *IEEE Transactions on Information Theory (TIT)*, 2022. [\[IEEE Communications Society & Information Theory Society Joint Paper Award\]](#)
 - [J7] **X. Chen**, S. Sarkar and M. H. Lotfi, [The Interplay of Competition and Cooperation among Service Providers \(Part II\)](#), *IEEE Transactions on Network Science and Engineering (TNSE)*, 2020.
 - [J8] **X. Chen**, S. Sarkar and M. H. Lotfi, [The Interplay of Competition and Cooperation among Service Providers \(Part I\)](#), *IEEE Transactions on Network Science and Engineering (TNSE)*, 2020.
 - [J9] **X. Chen**, Y. Liu, and Z. Wan, [Optimal Decision Making for Online and Offline Retailers under BOPS Mode](#), *ANZIAM Journal*, 2016.
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Conference

- [C1] **X. Chen**, P. Parag, R. Bhagat, Z. Liu, and S. El Rouayheb, [Random Walk Learning and the Pac-Man Attack](#), *IEEE ISIT*, 2026.
- [C2] **X. Chen**, N. NaderiAlizadeh, A. Ribeiro, and S. Saeedi Bidokhti, [Decentralized Learning Strategies for Estimation Error Minimization with Graph Neural Networks](#), *IEEE ICASSP*, 2026. [\[Full Paper\]](#)

- [C3] H. Nikpey, J. Kim, **X. Chen**, S. Sarkar, and S. Saeedi Bidokhti, [Group Testing with Correlation via Edge-Faulty Graphs](#), *IEEE ISIT*, 2022.
- [C4] **X. Chen**, R. Liu, S. Wang, and S. Saeedi-Bidokhti, [Timely Broadcasting in Erasure Networks: Age-Rate Tradeoffs](#), *IEEE ISIT*, 2021.
- [C5] **X. Chen**, X. Liao, and S. Saeedi-Bidokhti, [Real-Time Sampling and Estimation on Random Access Channels: Age of Information and Beyond](#), *IEEE INFOCOM*, 2021. [[IEEE INFOCOM Student Conference Award](#)]
- [C6] **X. Chen**, K. Gatsis, H. Hassani, and S. Saeedi-Bidokhti, [Age of Information in Random Access Channels](#), *IEEE ISIT*, 2020.
- [C7] **X. Chen** and S. Saeedi Bidokhti, [Benefits of Coding on Age of Information in Broadcast Networks](#), *IEEE ITW*, 2019.
- [C8] M. Fereydoonlian, **X. Chen**, H. Hassani, and S. Saeedi Bidokhti, [Non-Asymptotic Coded Slotted ALOHA](#), *IEEE ISIT*, 2019. [[IEEE ISIT Student Travel Grant](#)]

Professional Service

Editorial Board Positions

Guest Editor, Entropy, Special Issue: Goal-Oriented Communication: Freshness, Semantics, and Beyond, 2025.

Guest Editor, China Communications, Blue Ocean Forum, 2024.

Conference Organizations

Advisory TPC Member, IEEE INFOCOM, 2027.

TPC Member, WiOpt, 2026.

TPC Member, IEEE ICC: SAC Smart Grid Communications Track, 2024.

TPC Member, IEEE GLOBECOM Workshop: Information Freshness, Communications, Control, and Computing for Industrial IoT, 2020.

Technical Reviewers

IEEE Transactions on Information Theory (TIT),
 IEEE/ACM Transactions on Networking (ToN)
 IEEE Transactions on Mobile Computing (TMC)
 IEEE Journal on Selected Areas in Communications (JSAC)
 IEEE Transactions on Network Science and Engineering (TNSE)
 IEEE Transactions on Communications (TCOM)
 IEEE Transactions on Wireless Communications (TWC)
 IEEE Transactions on Artificial Intelligence (TAI)
 Journal of Computational and Applied Mathematics (JCAM)
 IEEE Wireless Communications Letters (WCL)
 IEEE Internet of Things Journal (IoT-J)
 IEEE Open Journal of the Communications Society (OJCOMS)

Presentations

Invited Talks

- [T1] *Self-Creating Random Walks for Decentralized Learning under Pac-Man Attacks*, Sun Yat-sen University - Zhuhai Campus, Zhuhai, China, 2026. [Host: Prof. Xuehe Wang]
- [T2] *Resilient and Efficient Random-Walk-Based Decentralized Learning*, University of Electronic Science and Technology of China, Chengdu, China, 2026. [Host: Prof. Yifan Chen]

- [T3] *Self-Creating Random Walks for Decentralized Learning under Pac-Man Attacks*, Singapore University of Technology and Design (SUTD), Virtual, 2026.
 - [T4] *Random Walk Learning and the Pac-Man Attack*, Joint Mathematics Meetings (JMM), Washington, D.C., UAS, 2026.
 - [T5] *Random Walk Learning and the Pac-Man Attack*, Rutgers University, New Brunswick, USA, 2025.
 - [T6] *Timely Requesting for Time-Critical Content Users in Decentralized F-RANs*, University of Pennsylvania, Philadelphia, USA, 2024.
 - [T7] *Timely Inferences for Updating Decisions in Networked Systems*, Central South University, Online (due to COVID-19), 2022. [\[Celebrating the 20th Anniversary of the School of Mathematics and Statistics\]](#)
 - [T8] *Benefits of Coding on Age of Information in Broadcast Networks*, Southwest University, Chongqing, China, 2019.
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Professional Talks

- [T1] Invited Program Talk, *Walk For Resiliency & Privacy: A Random Walk Framework for Learning at the Edge*, The 4th Resilient & Intelligent NextG Systems (RINGS) program PI meeting, Virtual, 2026.

Group

Current Master's Students

Yi Zhuang

Fall 2024 - present, Master's Student, UESTC
 Spring 2024, Undergraduate Thesis Student, Sichuan University
 Co-advised with Prof. Kun Yang

Taichun Yeh

Spring 2025 - present, Master's Student, Rutgers University

Current Undergraduate Students

Shunjie Bi

Summer 2026, Summer Intern, Imperial College London

Former Master's Students

Kai Li

Fall 2023 - Spring 2025, Master's Student, UESTC
 Co-advised with Prof. Kun Yang
 Next: Meituan (Industrial)

Former Undergraduate Students

Ruoqi Lu

Fall 2025 - Spring 2026, Undergraduate Research Assistant, Xidian University
 Co-advised with Prof. Kun Yang
 Next: Master's student at UESTC, China

Bohao Yang

Fall 2024 - Spring 2025, Undergraduate Thesis Student, UESTC
 Excellent Talents program for undergraduates
 Next: Master's student at Southeast University, China